On the Temporal Sparsity of Residual Connections: Periodic Messengers and DLA Invariance Extension 4

Theoretical Addendum

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Abstract

We investigate the effect of temporal sparsity on the Q-LINK architecture by applying the coherent residual pathway periodically (every k layers) rather than densely. We prove analytically that temporal sparsity fails to fracture the Dynamical Lie Algebra (DLA), meaning a k-periodic circuit ultimately explores the identical $\mathfrak{su}(2^{n+1})$ manifold and achieves the exact same asymptotic ensemble variance as the dense baseline. However, we show that periodicity acts as a mechanical throttle on operator spreading, fundamentally altering the finite-depth convergence to Haar saturation and offering a hyperparameter trade-off between expressive capacity and barren plateau onset.

1 DLA Invariance Under Temporal Sparsity

Let $\mathcal{G}_{opr}$ be the set of Lie generators corresponding to the data-register operation block, and $\mathcal{G}_{res}$ be the generators of the messenger collection and distribution blocks (e.g., $iX_m \otimes X_i$ and $iZ_m \otimes X_i$).

For the dense Q-LINK architecture ($k = 1$), every layer applies the full sequence, meaning the circuit accesses the full generator set: $\mathcal{G}_{dense} = \mathcal{G}_{opr} \cup \mathcal{G}_{res}$. For a k-periodic Q-LINK ($k > 1$), the circuit applies $\mathcal{G}_{opr}$ exclusively for $(k-1)$ layers, and the union $\mathcal{G}_{opr} \cup \mathcal{G}_{res}$ on every k-th layer.

The Dynamical Lie Algebra $\mathfrak{g}$ is defined as the closure of the available generators under nested commutators, $\mathfrak{g} = \langle \mathcal{G} \rangle_{Lie}$. Crucially, this algebraic closure is completely independent of the frequency or temporal sequence of the operations, provided the parameters are continuously independent and depth is unconstrained. Thus:

$$\mathfrak{g}_{periodic} = \langle \mathcal{G}_{opr} \cup \mathcal{G}_{res} \rangle_{Lie} = \langle \mathcal{G}_{dense} \rangle_{Lie} = \mathfrak{su}(2^{n+1}) \tag{1}$$

Theorem 1: Temporal sparsity cannot fracture the DLA. A k-periodic residual architecture generates the exact same continuous sub-manifold as a dense architecture.

2 Trace Invariance and Asymptotic Variance

Because the periodic architecture shares the same DLA ($\mathfrak{su}(2^{n+1})$) and identical initial state purity, the denominator of the Master Formula is unchanged. Furthermore, altering the layer frequency does not modify the structural definition of the extended Hilbert space ($\mathcal{H}_{tot} = 2^{n+1}$) nor the locality of the cost operator ($O \otimes I_m$).

Therefore, the Hilbert-Schmidt norm of the traceless cost operator remains invariant:

$$Tr(\tilde{V}^2_{periodic}) = Tr(\tilde{V}^2_{dense}) = \frac{2N}{n} \tag{2}$$

Applying the Master Formula at Haar saturation, the absolute gradient variance bound is mathematically identical for all $k \geq 1$. The asymptotic relative variance advantage over Vanilla remains strictly $\mathcal{R}(n) = \frac{n+1}{n}$.

3 The Finite-Depth Convergence Trade-off

While the asymptotic geometric bounds are invariant, periodicity profoundly affects the finite-depth circuit dynamics. Barren plateaus manifest when the circuit depth D is sufficient to form an exact unitary 2-design, saturating the Haar volume.

In a dense Q-LINK, the messenger qubit scrambles phase information globally across the data register at every layer, accelerating the expansion of the causal lightcone. In a k-periodic architecture, the effective path length of the residual connection is fractionated. Operator spreading is throttled, and the depth required to reach Haar saturation scales proportionally to $k \times D_{dense}$.

Conclusion: Applying the messenger periodically does not provide a structural geometric escape from the Haar volume penalty. Instead, periodicity serves as a dynamical hyperparameter. Increasing k artificially restricts expressibility at shallow depths, delaying t-design convergence and thereby extending the operable trajectory variance window before the global barren plateau ultimately sets in.